

T LEVEL

*Technical Qualification in Digital
Software Development*

Employer set project - Digital Software Development: Grade A Guide Exemplar Response

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Table of Contents

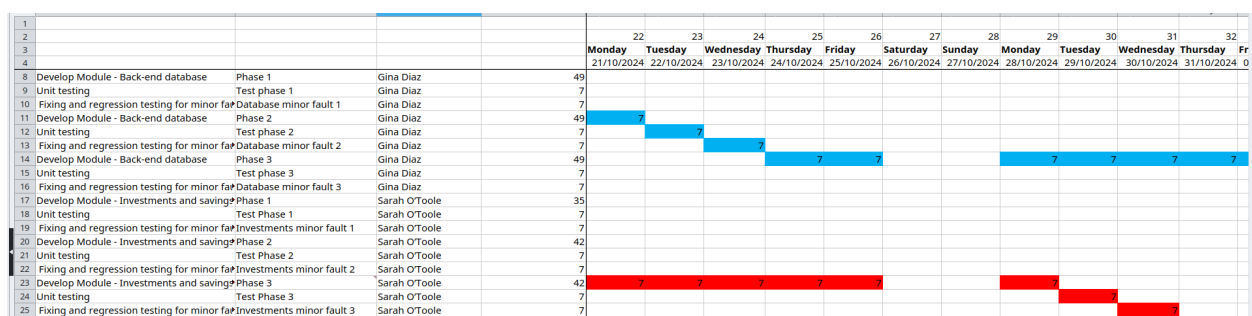
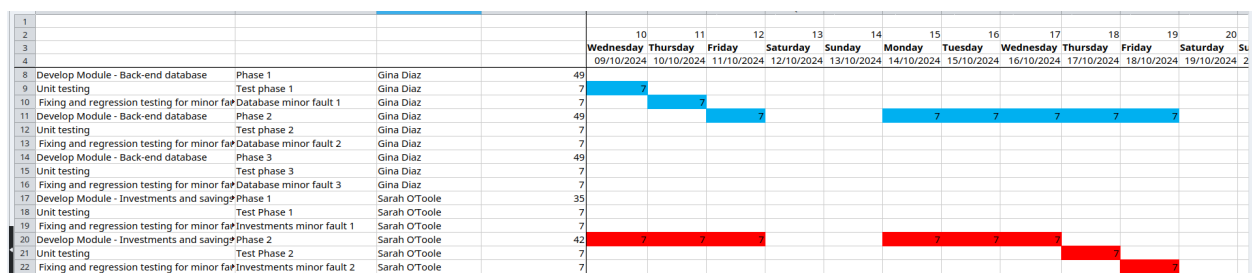
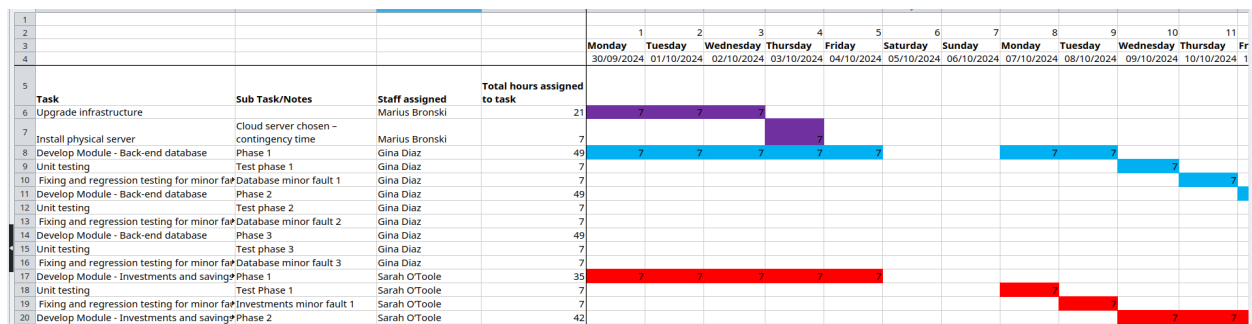
Task 1 - Example Student Response (A Grade)	3
Task 1 - Commentary	10
Task 2 - Example Student Response (A Grade)	12
Task 2 - Commentary	14
Task 3 - Example Student Response (A Grade)	15
Task 3 - Commentary	16
Task 4a - Example Student Response (A Grade)	17
Task 4a - Commentary	21
Task 4b - Example Student Response (A Grade)	22
Task 4b - Commentary	29

Task 1 - Example Student Response (A Grade)

The complete student response for this task is available in the accompanying download to this document. The evidence that relates to Task 1 for Grade A is:

- Task1_DSD_ProjectPlan_Grade_A_Example.xlsx
- Task1_DSD_Rationale_Grade_A_Example.pdf

Task 1 - Example Student Response (Project Plan - Gantt Chart)



				35	36	37	38	39	40	41	42	43	44	45
				Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday	Monday	Tuesday	Wednesday
				03/11/2024	04/11/2024	05/11/2024	06/11/2024	07/11/2024	08/11/2024	09/11/2024	10/11/2024	11/11/2024	12/11/2024	13/11/2024
14	Develop Module - Back-end database	Phase 3	Gina Diaz											
15	Unit testing	Test phase 3	Gina Diaz											
16	Fixing and regression testing for minor fa	Database minor fault 3	Gina Diaz											
17	Develop Module - Investments and saving	Phase 1	Sarah O'Toole											
18	Unit testing	Test Phase 1	Sarah O'Toole											
19	Fixing and regression testing for minor fa	Investments minor fault 1	Sarah O'Toole											
20	Develop Module - Investments and saving	Phase 2	Sarah O'Toole											
21	Unit testing	Test Phase 2	Sarah O'Toole											
22	Fixing and regression testing for minor fa	Investments minor fault 2	Sarah O'Toole											
23	Develop Module - Investments and saving	Phase 3	Sarah O'Toole											
24	Unit testing	Test Phase 3	Sarah O'Toole											
25	Fixing and regression testing for minor fa	Investments minor fault 3	Sarah O'Toole											
26	Integration testing	Investments and Database	Sarah O'Toole											
27	Fixing and regression testing for major fa	Predicted Major fault 1	Sarah O'Toole											
28	Fixing and regression testing for major fa	Predicted Major fault 1	Gina Diaz											
29	Deploy Modules	Deploy database and Investments	Terry Duran											
30	Develop Module - Data analytics	Phase 1	Gina Diaz											

				43	44	45	46	47	48	49	50	51	52	53
				Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday	Monday	Tuesday	Wednesday	Thursday
				11/11/2024	12/11/2024	13/11/2024	14/11/2024	15/11/2024	16/11/2024	17/11/2024	18/11/2024	19/11/2024	20/11/2024	21/11/2024
29	Deploy Modules	Deploy database and Investments	Terry Duran											
30	Develop Module - Data analytics	Phase 1	Gina Diaz											
31	Unit testing	Test Phase 1	Gina Diaz											
32	Fixing and regression testing for minor fa	Analytics Minor Fault 1	Gina Diaz											
33	Develop Module - Data analytics	Phase 2	Gina Diaz											
34	Unit testing	Test Phase 2	Gina Diaz											
35	Fixing and regression testing for minor fa	Analytics Minor Fault 2	Gina Diaz											
36	Develop Module - Data analytics	Phase 3	Gina Diaz											
37	Unit testing	Test Phase 3	Gina Diaz											
38	Fixing and regression testing for minor fa	Analytics Minor Fault 3	Gina Diaz											
39	Integration testing	Integrate analytics	Gina Diaz											
40	Deploy Modules	Deploy analytics	Gina Diaz											
41	Develop Module - Information feed	Phase 1	Ahad Shafiq											
42	Develop Module - Information feed	Phase 1	Terry Duran											
43	Unit testing	Test Phase 1	Ahad Shafiq											
44	Unit testing	Test Phase 1	Terry Duran											
45	Fixing and regression testing for minor fa	Information feed minor fault	Ahad Shafiq											
46	Fixing and regression testing for minor fa	Information feed minor fault	Terry Duran											

				53	54	55	56	57	58	59	60	61	62	63
				Thursday	Friday	Saturday	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
				21/11/2024	22/11/2024	23/11/2024	24/11/2024	25/11/2024	26/11/2024	27/11/2024	28/11/2024	29/11/2024	30/11/2024	01/12/2024
32	Fixing and regression testing for minor fa	Analytics Minor Fault 1	Gina Diaz											
33	Develop Module - Data analytics	Phase 2	Gina Diaz											
34	Unit testing	Test Phase 2	Gina Diaz											
35	Fixing and regression testing for minor fa	Analytics Minor Fault 2	Gina Diaz											
36	Develop Module - Data analytics	Phase 3	Gina Diaz											
37	Unit testing	Test Phase 3	Gina Diaz											
38	Fixing and regression testing for minor fa	Analytics Minor Fault 3	Gina Diaz											
39	Integration testing	Integrate analytics	Gina Diaz											
40	Deploy Modules	Deploy analytics	Gina Diaz											
41	Develop Module - Information feed	Phase 1	Ahad Shafiq											
42	Develop Module - Information feed	Phase 1	Terry Duran											
43	Unit testing	Test Phase 1	Ahad Shafiq											
44	Unit testing	Test Phase 1	Terry Duran											
45	Fixing and regression testing for minor fa	Information feed minor fault	Ahad Shafiq											
46	Fixing and regression testing for minor fa	Information feed minor fault	Terry Duran											
47	Develop Module - Information feed	Phase 2	Ahad Shafiq											
48	Develop Module - Information feed	Phase 2	Terry Duran											
49	Unit testing	Test Phase 2	Ahad Shafiq											
50	Unit testing	Test Phase 2	Terry Duran											

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83	Develop Module - User interface	Phase 2	Ahad Shafiq	21											
84	Unit testing	UI test Phase 2	Sarah O'Toole	7	7										
85	Unit testing	UI Test Phase 2	Ahad Shafiq	7	7										
86	Fixing and regression testing for minor fa	UI Minor Fault 2	Sarah O'Toole	7											
87	Fixing and regression testing for minor fa	UI Minor Fault 2	Ahad Shafiq	7											
88	Develop Module - User interface	Phase 3	Sarah O'Toole	28											
89	Develop Module - User interface	Phase 3	Ahad Shafiq	28											
90	Unit testing	UI test Phase 3	Sarah O'Toole	7											
91	Unit testing	UI Test Phase 3	Ahad Shafiq	7											
92	Fixing and regression testing for minor fa	UI Minor Fault 3	Sarah O'Toole	7											
93	Fixing and regression testing for minor fa	UI Minor Fault 3	Ahad Shafiq	7											
94	Integration testing	integrate UI	Sarah O'Toole	14											
95	Deploy Modules	Deploy UI	Sarah O'Toole	7											
96	Fixing and regression testing for major fa	Predicted major fault 3	Sarah O'Toole	14											
97	User Training		Sarah O'Toole	21											
98	User/acceptance testing		Terry Duran	35											
99	Create a Test plan		Terry Duran	14											
100		Project manager		588	7	7	7	7	7	7		7	7	7	7

[illegible]

Task 1 - Example Student Response (Project Plan - Cost plan)

[illegible]

Task 1 - Example Student Response (Rationale)

As a result of my planning of the time scales, tasks and costs of the project, I believe that although it is possible to complete the project in the requested time frame of 17 weeks, the timeline will still be very tight and would only make the deadline by a couple of days

Additionally, the project would be affordable and continue to return good levels of profit over the next three years.

In this report I will highlight some of the key decisions I made and how these influenced my final findings. I will also provide a rationale for my decisions and cover potential impacts on the project.

Order of tasks considering dependencies and prerequisites

One of the first set of tasks I decided to schedule was the upgrading of the infrastructure and installing the physical server. I scheduled this first as I think it is vital that we get the target hardware completed so that modules can be deployed and tested on the systems that they will be used on as soon as possible. Also, this means that the Network Engineer (Marius) can be freed up to work on other projects for the company as soon as this task is completed.

The first module I scheduled for development was the back-end database, many of the other modules are likely to either draw data from this, or need to write data to it, so it is important that this is completed early in the project.

In order to ensure I met the very tight timelines, I did have to schedule one module to be developed at the same time as the back-end database, specifically the investments and savings module. While this is likely to need part of the database, it would probably also need external data, so it would be possible to complete some of the functionality before the database is complete, and as more of the database is developed, they could integrate that functionality. I scheduled integration testing for these two modules to happen at the same time, so that Gina and Sarah could work closely to ensure that the two modules share data as needed and any issues are resolved before the rest of the project continues.

Because of the difficulty of developing these two modules at the same time, I anticipate that one of the project's three major faults could happen during this phase, therefore I have added this to my plan. For the other predicted faults, I have chosen to spread these out across the whole project, as it is difficult to predict exactly when they will occur. However, I have scheduled them after integration of modules happens, as this is firstly a possible time when issues could occur but also allows me to anticipate possible delays at different stages of the project and therefore be able to stick roughly to the subsequent times if an issue does occur.

For each of the modules I chose to take an agile approach. There are three predicted minor faults for each module, which typically would be revealed through testing, so I have chosen to split each module into three “sprints” allowing 7 hours of testing at each stage (meeting the total for each module of 21 hours of testing).

Allocation of tasks to staff

I have allocated staff based on their skill and experience, as provided in the task information, and I have tried to ensure that the chosen modules are appropriate for the staff member. As such Sarah (the Senior Developer) was allocated to the Investment and savings module. Because this is both important, and as I have mentioned earlier, it is a difficult task to complete while the database is still also being developed, I thought it would be sensible to allocate the most experienced software engineer to this task.

Gina is a Data specialist, so I have mainly only used her for the back-end database module, however I have also allocated her to work on the fixing and regression testing of any major faults along with Sarah. I have allocated both staff members to this as the system is heavily reliant on data so a major fault is likely to involve a number of software and data components, therefore using the most experienced Software Engineer and a data specialist should allow us to tackle any problems effectively.

The junior developers I have always allocated to work together on the same task. I have

chosen to do this because Ahad is very inexperienced having only just graduated university, so I would not expect him to work independently on modules, especially on such a high-risk system.

Potential benefits and risks

One particular concern that I noticed was that there is no testing task scheduled for hardware testing. I would assume that the engineer should test the infrastructure and server as they go, but not having it specifically scheduled could lead to specific testing not being carried out resulting in issues later.

Choosing to produce the investments module at the same time as the database is very risky because if part of the database that is needed for the investments module isn't completed early in the process, this would delay the whole project. However, the two staff I have allocated are very experienced and as such should be aware of such potential issues, which can be prevented or planned for through communication between the two staff members.

The agile method I have chosen is a definite benefit; by breaking the development of each module into shorter sprints and pairing these with testing and fixing of minor faults, the impact of a fault should be lessened compared to doing all the testing in one go. As such this should result in fewer unforeseen issues arising and allow us to stick to the predicted timings.

I chose to use a cloud server for the company, although it is more expensive over the course of three years, the initial set up cost is lower, and it would provide better scalability for a growing business.

I did notice that there was no time allocation predicted for the cloud server, I assume that because it doesn't require any physical hardware and they can be set up quite quickly that this would form part of the Module development and deployment tasks. However, I think this is very risky indeed and there should be some specific times allocated in the plan for it. I have therefore allocated the same time as the physical server for Marius to work on this task.

Given the sensitive nature of the data (people's financial data) hosting on the cloud does present security risks, so a reliable cloud provider would need to be chosen

The impact of planning decisions on the final timings and costs of the project

My planning suggests that the project would be able to be completed on time. However, to achieve this I have not allocated any contingency time into the plan. As such any delay would result in the project going over the deadline.

I chose the cloud server due to its reduced initial set up costs. Although this is more expensive in the long run, the trade off between scalability and the fact that the

underlying system hardware would be maintained by a third party, resulting in greater reliability, is worth paying extra for.

Choosing to pair the two junior software engineers on all the tasks they do does add a little more expense to the project, but I think this is a good investment as it allows the two workers to team code and check each other's work resulting in much higher quality products.

Overall, my decisions make only a minimal impact on the overall affordability of the project, the project is very affordable based on the company's current income and would result in a significant increase in profit over the next three years.

Task 1 - Commentary

Gantt Chart

The student has shown a highly logical and effective approach to project planning. The student has used an agile mode to break longer modules down into shorter development phases, with testing sensibly spread throughout the project and placed at sensible points in conjunction with the fixing and regression testing of potential issues.

There is appropriate use of parallel tasks and concurrent tasks, starting with the infrastructure upgrade and the development of core module (Back-end Database). The student has chosen to start the development of the Investments and Savings module at the same time, which would have its benefits and drawbacks, however it can be seen in the student's rationale that they have clearly reasoned this through.

Staff allocation is appropriate for the most part with staff being assigned tasks that suit their skill and experience profile. The very junior member of staff has been teamed up with a more experienced member of staff throughout the project, which is a sensible approach when a team contains such an inexperienced member of staff.

The student's effective planning has enabled them to correctly identify that it is possible to meet the requested deadline, although there is very little room for flexibility.

There are some improvements that could be made in Task 1, for example in their plan, it may have been sensible, later in the project to allocate the Junior Software engineers to separate but minor tasks, for example they could be used to carry out some testing activities, while the more senior engineers are developing modules or fixing other issues. Also, the project plan does not include any slack/contingency time, this is quite a long project so it would be sensible to assume that some drift would occur, and this should be planned for.

Project costings

The student has appropriately expanded the costs sheet from the given template, providing an accurate calculation of ongoing costs and costs that would only apply to the implementation of the project.

They have considered the impact of their planning on the current year and provided an accurate estimation of the ongoing financial impact of the project for the next three years.

The student has also correctly accounted for the cost/pay of the Project Manager. As you will see from the example file, they have included the Project Manager in the Gantt chart and used this to correctly calculate the number of hours worked, and subsequently the number of days and correct pay.

Rationale

The student has provided meaningful and insightful reasons for the planning decisions they have made. The reasons provided consider how the decisions made would impact on other tasks and projects as a whole, providing an insight into the thought process of the student, and demonstrating a thorough understating of the context, and how project planning skills can be applied, in a digital infrastructure project.

For example, they have provided a clear identification of why they have chosen (correctly) to develop the Back-end Database first, and why they have also chosen to schedule the Investments and Savings module at the same time. They have provided a clear and balanced explanation of their thought process considering the risks and benefits of this approach, providing clear insight into their thought making process.

The student has provided clear rationales throughout, for example providing justification for the way they have organised their teams, and how they have allocated some of the tasks. They have ensured that they have focused their response on “notable” decisions. For example, they have explained why they have placed the major faults at key points in the project (e.g. after integration testing).

They have shown an understanding of, and a rationale for, how their decisions impact on other aspects of the project planning. For example, they have explored the potential benefits and risks of always scheduling the two junior staff members to work on the same task at the same time and explored this both in relation to quality of outcomes and project costs.

To further improve their rationale there could be greater exploration of why they chose an agile model, the benefits this provides in the given project, and how they applied it to this scenario (i.e. they could go into greater depth about how they split and allocated the testing phases).

Overall, the student’s work demonstrates a significant number of excellent aspects such as:

- The student is able to comprehensively analyse a problem, showing a detailed problem and perceptive consideration of the factors that apply to effectively plan a digital solution
- Effective use of English, maths and digital skills to respond to a brief (for example appropriate maths skills to calculate the costs of the project).

Task 2 - Example Student Response (A Grade)

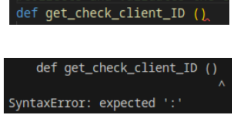
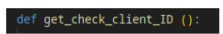
The complete student response for this task is available in the accompanying download to this document. The evidence that relates to Task 2 for Grade A is:

- Task2_DSD_Test_Log_Grade_A_Example.pdf
- Task2_DSD_Code_Grade_A_Example

Task 2 - Example Student Response (Test Log)

The following screenshots provide only extracted examples of the work. The full solution can be found in the accompanying download. Please take the time to read through the complete solution alongside the commentary.

Test 2 - Test Log

Description of test	Test data to be used (if required)	Expected outcome	Actual outcome	Comments and intended actions
Visual Debug and initial run	N/A	Code should run or provide output highlighting any syntax errors that would cause it not to run.	Line 2 - a colon ins missing in the function definition. Which will prevent it running. 	Colon added. 
Run program to confirm fix on line 2 works	N/A	Code should progress to line 2 and highlight any other syntax errors.	Error in name when calling main functional 	Name of function call corrected. 
Run program to confirm fix on line 148 works	N/A	Code should call the main function and prompt the user to write enter their name.	Works as expected 	Initial syntax errors fixed. I will move on to testing specific functionality

Description of test	Test data to be used (if required)	Expected outcome	Actual outcome	Comments and intended actions
Check Program correctly validates the length of the client ID number at the boundary of the logical test	123456789	Too few characters should be rejected	Did not work as expected - Number was accepted Please enter the name of the client: Ian Please enter the Client_ID number: 123456789 Client_ID accepted Please enter the amount to be invested per year:	Logic of the length check get_check_client_ID function is incorrect: <pre>def get_check_client_ID(clientID): if len(clientID) > 10: print('Client_ID must be 10 characters long')</pre> only checking if the length is more than 10 rather than ensuring it is exactly 10. Changed logic to != 10. <pre>if len(clientID) != 10:</pre>
Check change of logic has worked. Check Program correctly validates the length of the client ID number at the boundary of the logical test	123456789	Too few characters should be rejected	Works as expected - ID was rejected Please enter the Client_ID number: 123456789 Client_ID must be 10 characters long Please enter the Client_ID number:	Confirm logic works by testing upper boundary
Check change of logic has worked. Check Program correctly validates the length of the client ID number at the boundary of the logical test	12345678910	Too many characters should be rejected	Works as expected - ID was rejected Please enter the Client_ID number: 12345678910 Client_ID must be 10 characters long Please enter the Client_ID number:	Issue resolved

Description of test	Test data to be used (if required)	Expected outcome	Actual outcome	Comments and intended actions
Check standard savings plan option provide correct outputs	Amount invested per year = 1000	Maximum value = 5371.71 Fees for max = 39.67 Minimum value = 5182.89 Fees for min = 38.55	Maximum and minimum value are incorrect Fees are also incorrect. ***** Investment Summary ***** Client name: Jane Client ID: 123456789 ***** Savings plan summary ***** Initial Investment: £ 1000.0 If the savings plan performs at the highest rate after 5 years you can expect: Value of investment: £ 6245.81 Fees paid: £ 39.67 Profit made: £ 5245.81 If the savings plan performs at the lowest rate after 5 years you can expect: Value of investment: £ 4121.43 Fees paid: £ 38.55 Profit made: £ 3121.43	Maximum and minimum should be at least 5000. As it is 1000 per year. This suggests that it is not calculating the correct number of years. The range that controls the for loop is incorrect. A range of 1 to 5 would only loop 4 times. <pre>def savings_plan (InvestmentAmount): maximumValue = 0 minimumValue = 0 feesMax = 0 feesMin = 0 for i in range (1,5):</pre> Changed the lower number to 0 as counting for loops and indexes starts at 0. <pre>for i in range (0,5):</pre>
Check updated loop corrects the calculation Check standard savings plan option provide correct outputs	Amount invested per year = 1000	Maximum value = 5371.71 Fees for max = 39.67 Minimum value = 5182.89 Fees for min = 38.55	Calculations re now correct. ***** Investment Summary ***** Client name: Jane Client ID: 123456789 ***** Savings plan summary ***** Initial Investment: £ 1000.0 If the savings plan performs at the highest rate after 5 years you can expect: Value of investment: £ 5371.73 Fees paid: £ 39.67 Profit made: £ 5332.06 If the savings plan performs at the lowest rate after 5 years you can expect: Value of investment: £ 5182.91 Fees paid: £ 38.55 Profit made: £ 5144.36	Basic savings option tested and issues resolved.

Task 2 - Commentary

The Test log demonstrates a systematic and effective approach to diagnosing the issues and applies a series of effective solutions that results in a final solution that resolves all issues with minimal further actions required.

For example, the student uses a series of tests to establish that the code effectively validates the length of the customer ID number and includes manual calculations to check that the outputs for the standard investments calculations are accurate and uses these to effectively identify and then correct the problem.

The code file provided shows a consistent approach to that of their test log and demonstrates that they have drawn on the set task information, and an understanding of coding conventions to ensure the solution meets the task requirements.

There are, however, some minor improvements that could be made. For example, they could include more evidence of having retested issues after they are “resolved”, for example testing the calculations in the different functions with a range of different values, to ensure outputs are correct and consistent.

The student has also not addressed the outputs of calculations to the screen to ensure that they output as currency to 2 decimal places. However, this is a very minor issue, and the student has rightly prioritised accuracy of calculations and the functionality of the program.

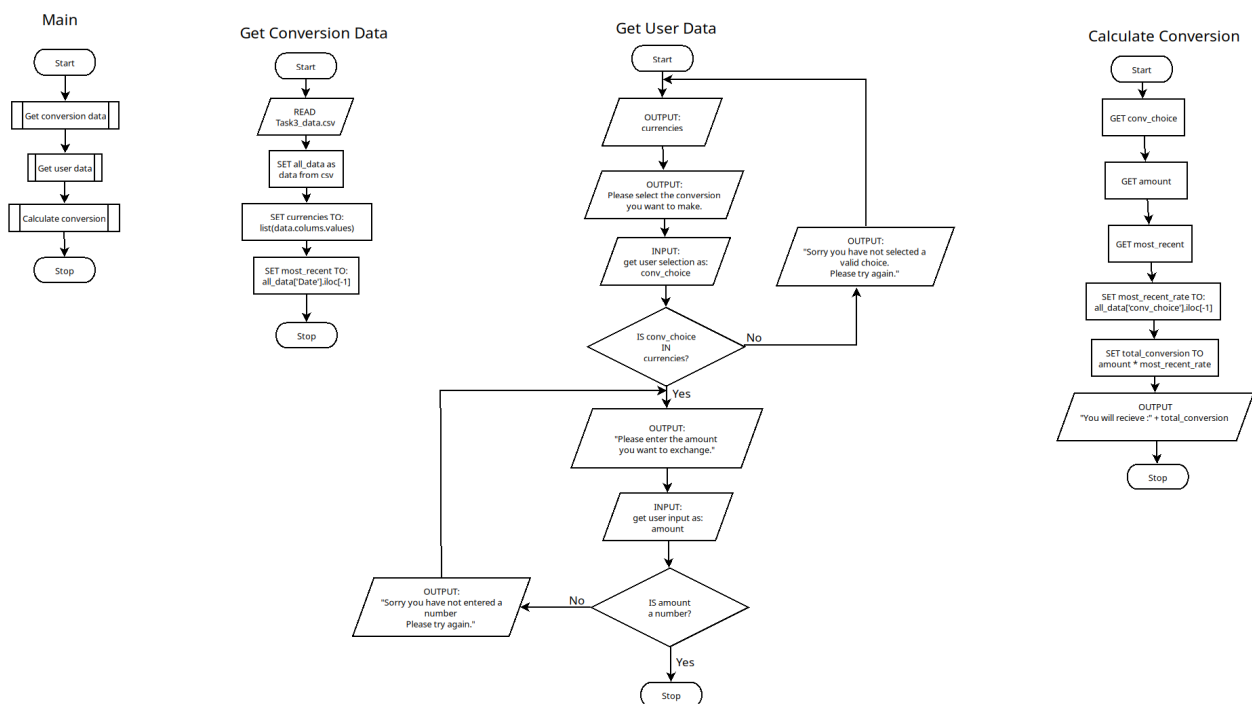
Overall, the student’s work demonstrates a significant number of excellent aspects which would contribute to meeting the Grade A descriptor such as:

- The student has selected a range of tests that will check both the basic functionality of the program and the logic and calculations.
- They use an effective range of test data including boundary testing and manual calculations to ensure all aspects of the program are performing as expected.
- The student has shown the errors they encountered and given a clear and detailed explanation of how they solved the problems.
- The student has selected and applied appropriate knowledge and skills to correct the given code.

Task 3 - Example Student Response (A Grade)

The complete student response for this task is available in the accompanying download to this document. The evidence that relates to Task 3 for Grade A is:

- Task3_DSD_Flowcharts_Grade_A_Example.pdf



Task 3 - Commentary

The student has demonstrated detailed decomposition of the requirements. They have provided details of appropriate Inputs, processes and outputs that would effectively address the requirements of the task. For example, the proposed solution effectively breaks down the main program into a number of sub-processes, each focusing on a key problem. The design then provides an accurate and detailed breakdown of each of these problems as its own flow chart.

The processes and logic of the flowcharts are accurate and would result in correct and consistent outcomes if implemented as a software solution.

The layout of the flowcharts is logical, and for the most part it is clear how they would implement their solution.

Overall, this is a highly effective design which could quite easily be used by a third-party developer to implement the intended solution. However, minor improvements could be made. Although presenting the design as a flowchart would help, a non-technical audience (e.g. the management team of the RBSX) to see the program flow, the content of the flowchart often focuses on the use of specific syntax for the Pandas Library, as such it would be difficult to understand by anyone not familiar with that specific library.

This could be mitigated by:

- either combining the use of flowcharts for higher level processes with the use of Pseudocode to describe the more technical code aspects
- OR
- adding further annotations alongside the flowcharts to explain any technical aspects or any aspects that are not readily obvious (e.g. what exactly will be output to the screen in the “OUTPUT currencies” box).

Overall, the student’s work demonstrates a significant number of excellent aspects which would contribute to meeting the Grade A descriptor such as:

- the student was able to analyse the problem by showing a detailed consideration of the factors that apply to effectively plan a digital solution
- the design is supported by effective use of digital skills (specifically flow chart conventions) to communicate technical information.

Task 4a - Example Student Response (A Grade)

The complete student response for this task is available in the accompanying download to this document. The evidence that relates to Task 4a for Grade A is:

- Task4a_code_Grade_A_Example.py

Task 4a - Example Student Response

The following screenshots provide only extracted examples of the work. The full solution can be found in the accompanying download. Please take the time to read through the complete solution alongside the commentary.

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 def main_menu():
5     flag = True
6
7     while flag:
8         print("#####")
9         print("Welcome to the RBSX currency analysis tool")
10        print("Please select an option")
11        print("1. Calculate currency conversion")
12        print("2. Compare GBP to other currencies")
13        print("3. Currency performance analysis")
14        print("")
15        print("#####")
16
17        menu_choice = input("Please enter the number of your choice (1-3): ")
18
19        try:
20            int(menu_choice)
21        except:
22            print("Sorry, you did not enter a valid choice")
23            flag = True
24        else:
25            if int(menu_choice) < 1 or int(menu_choice) > 3:
26                print("Sorry, you did not enter a valid choice")
27                flag = True
28            else:
29                return menu_choice
30
```

```
26 #Compares GBP to the selected currencey
27 #provides a grpah f the data over the slected timeframe
28 #caluclates teh average exchange rate in the given time frame
29 def process_value_data(range,currency_header):
30
31     main_df = pd.read_csv("Task4a_RBSX_data.csv")
32
33
34     if range == "1":
35         gbp_df = main_df[["Date",currency_header]]
36         seven_day_gbp = gbp_df.iloc[-7:]
37         seven_day_avg = round(seven_day_gbp[currency_header].mean(),3)
38         seven_day_gbp.plot(x="Date", y = currency_header)
39         plt.show()
40         print("GBP to {} for the last 7 days".format(currency_header))
41         print(seven_day_gbp)
42         print("")
43         print("The aveage convesion rate for the last 7 days is: ", seven_day_avg )
44
45     elif range == 2:
46         gbp_df = main_df[["Date", currency_header]]
47         fourteen_day_gbp = gbp_df.iloc[-14:]
48         fourteen_day_avg = round(fourteen_day_gbp[currency_header].mean(),3)
49         fourteen_day_gbp.plot(x="Date", y = currency_header)
50         plt.show()
51         print("GBP to USD last 14 days")
52         print(fourteen_day_gbp)
53         print("")
54         print("The aveage convesion rate for the last 14 days is: ", fourteen_day_avg )
55
56     else:
57         gbp_df = main_df[["Date", currency_header]]
58         thirty_day_gbp = gbp_df.iloc[-30:]
59         thirty_day_avg = round(thirty_day_gbp[currency_header].mean(),3)
60         thirty_day_gbp.plot(x="Date", y = currency_header)
61         plt.show()
62         print("GBP to USD last 30 days")
63         print(thirty_day_gbp)
64         print("")
65         print("The aveage convesion rate for the last 30 days is: ", thirty_day_avg )
```

```

67 def process_performance_data(range,chosen_currency):
68
69     main_df = pd.read_csv("Task4a_RBSX_data.csv")
70
71
72     if range == "1":
73         currency_df = main_df[["Date",chosen_currency]]
74         seven_day_perf = currency_df.iloc[-7:]
75         start = seven_day_perf[chosen_currency].iloc[1]
76         end = seven_day_perf[chosen_currency].iloc[-1]
77         variation = end - start
78         print("The starting value of this currency compared to GBP is:",start)
79         print("The final value of this currency compared to GBP is:",end)
80         if variation > 0:
81             print("This currency has increased in value over the last 7 days")
82         elif variation < 0:
83             print("This currency has decreased in value over the last 7 days")
84         else:
85             print("This currency has not changed in value over the last 7 days")
86
87     elif range == "2":
88         currency_df = main_df[["Date",chosen_currency]]
89         seven_day_perf = currency_df.iloc[-14:]
90         start = seven_day_perf[chosen_currency].iloc[1]
91         end = seven_day_perf[chosen_currency].iloc[-1]
92         variation = end - start
93         print("The starting value of this currency compared to GBP is:",start)
94         print("The final value of this currency compared to GBP is:",end)
95         if variation > 0:
96             print("This currency has increased in value over the last 14 days")
97         elif variation < 0:
98             print("This currency has decreased in value over the last 14 days")
99         else:
100             print("This currency has not changed in value over the last 14 days")
101
102     else:
103         currency_df = main_df[["Date",chosen_currency]]
104         seven_day_perf = currency_df.iloc[-30:]
105         start = seven_day_perf[chosen_currency].iloc[1]
106         end = seven_day_perf[chosen_currency].iloc[-1]
107         variation = end - start
108         print("The starting value of this currency compared to GBP is:",start)
109         print("The final value of this currency compared to GBP is:",end)
110         if variation > 0:
111             print("This currency has increased in value over the last 30 days")
112         elif variation < 0:
113             print("This currency has decreased in value over the last 30 days")
114         else:
115             print("This currency has not changed in value over the last 30 days")

```



```
16 def main():
17     main_choice = main_menu()
18
19     if main_choice == "1":
20         menu_choice = menu1()
21         currency = get_currency(menu_choice)
22         conversion_rate = get_conversion_rate(currency)
23         conversion_amount = float(get_amount_to_convert(currency))
24         perform_conversion(conversion_amount, conversion_rate, currency)
25     elif main_choice == "2":
26
27         currency_header = gbp_comparison()
28         range = get_compare_range()
29         process_value_data(range, currency_header)
30
31     elif main_choice == "3":
32         chosen_currency = currency_performance()
33         range = get_compare_range()
34         process_performance_data(range, chosen_currency)
35
36 main()
```

Task 4a - Commentary

The student has expanded the given code effectively and addresses all of the requirements listed in the set task information

The code effectively provides functionality that allows the user to select a currency and compare that to the value of GBP. The code provides the user with options for three different time ranges and effectively extracts data from the csv file to show the performance of the currency.

The code also addresses the second requirement by calculating the increase or decrease in the value of a currency over a selected time frame.

The code is well organised, making effective use of functions to separate code, ensuing secure coding practices (data is only extracted and used within the function) and the maintainability is provided through logical structure and meaningful naming conventions.

The program outputs are, for the most part, useful and meaningful providing the user with good experience and helpful and accurate information. For example, they have provided performance data in both numerical/tabulated format and in the form of a line graph.

There are some small areas of improvement that could be made, however. For example:

- the titles and layout of the line graphs could be improved - the layout and number of data points on the X-axis are most problematic with the 30-day data.
- greater flexibility could be provided for the user, for example the student could provide the capability to select a custom data/time range rather than just the last 7,14 and 30 days

Overall, the student's work demonstrates a significant number of excellent aspects which would contribute to meeting the Grade A descriptor such as:

- The student was able to comprehensively analyse the problem and effectively solve both straight forward and complex aspects of the problem.
- They have effectively applied knowledge, skills and processes to develop software.
- They have appropriately selected parts of the given code to reuse and combined it effectively with code that they have written themselves.

Task 4b - Example Student Response (A Grade)

The complete student response for this task is available in the accompanying download to this document. The evidence that relates to Task 4b for Grade A is:

- Task4b_DSD_Review_Grade_A_Example.pdf

Task 4b - Code review

Review of how my code met the system requirements.

Overall, I believe that I have produced a solution that meets all the system requirements to a good standard.

The first thing I did was to add an initial menu that provides the user with three options, one for each of the main system requirements given in the task.

In this section I ensured the code was consistent and robust by refactoring some of the code provided in the original code file.

```
def main_menu():
    flag = True

    while flag:
        print("#####")
        print("Welcome to the RBSX currency analysis tool")
        print("Please select an option")
        print("1. Calculate currency conversion")
        print("2. Compare GBP to other currencies")
        print("3. Currency performance analysis")
        print("")
        print("#####")

        menu_choice = input("Please enter the number of your choice (1-3): ")

        try:
            int(menu_choice)
        except:
            print("Sorry, you did not enter a valid choice")
            flag = True
        else:
            if int(menu_choice) < 1 or int(menu_choice) > 3:
                print("Sorry, you did not neter a valid choice")
                flag = True
            else:
                return menu_choice
```

I adapted the range of the input check from the original code so that it provided appropriate validation for the user input.

Next, I restructured some of the code to fit with the approach I took with the initial menu.

The lines of code that I have grouped under main_choice 1 were originally spread out throughout the code base and were manipulated in the global space. I chose instead to keep them contained within the main() function and pass variables between them, as such my code follows secure coding practices helping me meet the requirement for “must be secure”

```
def main():  
    main_choice = main_menu()  
  
    if main_choice == "1":  
        menu_choice = menu1()  
        currency = get_currency(menu_choice)  
        conversion_rate = get_conversion_rate(currency)  
        conversion_amount = float(get_amount_to_convert(currency))  
        perform_conversion(conversion_amount, conversion_rate, currency)
```

The first additional requirement I addressed was showing *trends and patterns the value of the pound (GBP) compared to other currencies*.

To meet this requirement, I organised my code into three main processes which are contained within functions.

Firstly, the code gets the user's choice of currency that they wish to view and returns a string that can be used to identify the hearer in the dataframe.

```
def currency_performance():
    flag = True

    while flag:
        print("Please select the currency you wish to see the performance of:")
        print("1. Euros (EUR)")
        print("2. Australian Dollars (AUD)")
        print("3. Japanese Yen (JPY)")
        print("#####")

        compare_choice = input("Please enter the number of your choice (1-3): ")

        try:
            int(compare_choice)
        except:
            print("Sorry, you did not enter a valid choice")
            flag = True
        else:
            if int(compare_choice) < 1 or int(compare_choice) > 3:
                print("Sorry, you did not enter a valid choice")
                flag = True
            else:
                flag = False

        if compare_choice == "1":
            currency_header = "EUR - GBP"
        elif compare_choice == "2":
            currency_header = "AUD - GBP"
        else:
            currency_header = "JPY - GBP"

    return currency_header
```

I think that this code is a little inefficient, as the headings are hard coded. I did start to develop a function that extracted the header row from the CSV, which would make the code more scalable, however I was not sure I could get it working fully in the time I had, so I chose instead to prioritise functional over efficient code.

This is the start of the alternative code I was going to use:

```
#extracts the headers from the csv file
def get_currency_headers():
    data = pd.read_csv("Task4a_RBSX_data.csv")
    currencies = list(data.columns.values)
    currencies.pop(0)
    return currencies
```

The second part of the code, for this gets the user to choose the time frame that they wish to see the comparison for.

I made the code easy to use by providing three options, for which the user only has to enter the relevant number, this reduces the chance of user error, and the simplicity of this ensures a good user experience.

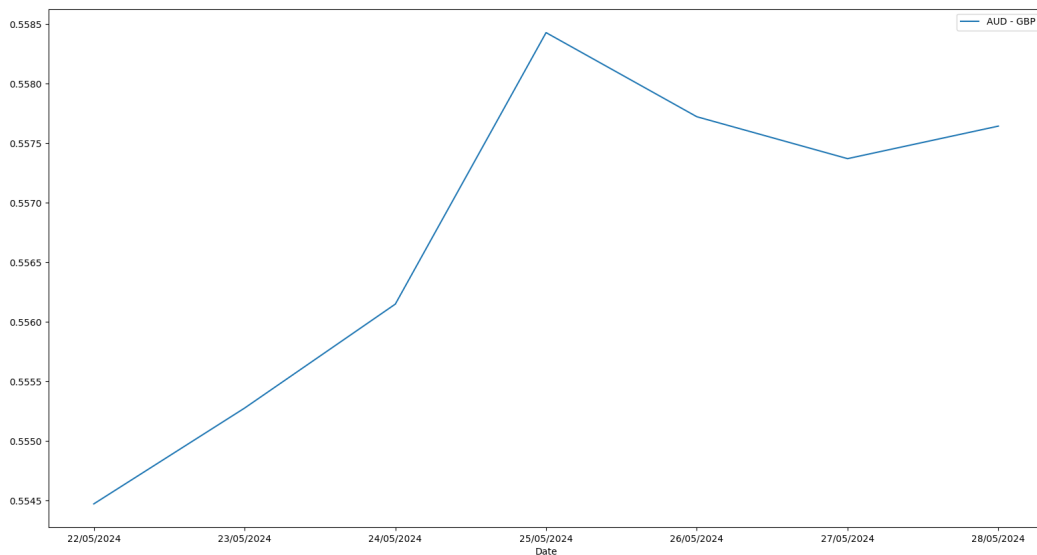
```
def get_compare_range():  
    flag = True  
  
    while flag:  
        print("#####")  
        print("##### Historical data #####")  
        print("#####")  
        print("Please choose a time fram for your comparison")  
        print("1. Last 7 days")  
        print("2. Last 14 days")  
        print("3. Last 30 days")  
        print("#####")  
  
        range_choice = input("Please enter the number of your choice (1-3): ")  
  
        try:  
            int(range_choice)  
        except:  
            print("Sorry, you did not enter a valid choice")  
            flag = True  
        else:  
            if int(range_choice) < 1 or int(range_choice) > 3:  
                print("Sorry, you did not neter a valid choice")  
                flag = True  
            else:  
                return range_choice
```

I chose to provide three predefined time options, 7 days, 14 days and 30 days which provides the last 7,14 and 30 entries in the data file. I chose to do this because the first part of the requirements was to use “most up to date figures”. Although this was only stated in the first requirement and related to the code that was provided, I thought using more recent data is more relevant to a user’s needs.

The final part of the is functionality, uses the iloc method from pandas to extract the required data from the initial data frame. I chose to output this data in both graphical and tabulated form to meet a range of user needs.

I also provided an average value of the conversion rate over the given time period.

```
def process_value_data(range,currency_header):  
  
    main_df = pd.read_csv("Task4a_RBSX_data.csv")  
  
    if range == "1":  
        gbp_df = main_df[["Date",currency_header]]  
        seven_day_gbp = gbp_df.iloc[-7:]  
        seven_day_avg = round(seven_day_gbp[currency_header].mean(),3)  
        seven_day_gbp.plot(x="Date", y = currency_header)  
        plt.show()  
        print("GBP to {} for the last 7 days".format(currency_header))  
        print(seven_day_gbp)  
        print("")  
        print("The average conversion rate for the last 7 days is: ", seven_day_avg )
```




```
##### Historical data #####
#####
Please choose a time frame for your comparison
1. Last 7 days
2. Last 14 days
3. Last 30 days
#####
Please enter the number of your choice (1-3): 1
GBP to AUD - GBP for the last 7 days
      Date  AUD - GBP
79 22/05/2024  0.554472
80 23/05/2024  0.555276
81 24/05/2024  0.556149
82 25/05/2024  0.558428
83 26/05/2024  0.557722
84 27/05/2024  0.557370
85 28/05/2024  0.557643

The average conversion rate for the last 7 days is: 0.557
```

The second first additional requirement I addressed was showing *trends and patterns for the performance (e.g. increase/decrease in value) of a selected currency*.

As with the first additional requirement, I broke the code into three main processes. The first two were very similar in that they were focused on getting the user's choices.

The final part then extracts the data for the chosen currency over the chosen time frame.

In order to show how the currency performed, I extracted the first and last value from the subset of data and then subtracted the initial value from the final value to see if there had been an increase or not.

```
def process_performance_data(range,chosen_currency):

    main_df = pd.read_csv("Task4a_RBSX_data.csv")

    if range == "1":
        currency_df = main_df[["Date",chosen_currency]]
        seven_day_perf = currency_df.iloc[-7:]
        start = seven_day_perf[chosen_currency].iloc[1]
        end = seven_day_perf[chosen_currency].iloc[-1]
        variation = end - start
        print("The starting value of this currency compared to GBP is:",start)
        print("The final value of this currency compared to GBP is:",end)
        if variation > 0:
            print("This currency has increased in value over the last 7 days")
        elif variation < 0:
            print("This currency has decreased in value over the last 7 days")
        else:
            print("This currency has not changed in value over the last 7 days")
```

How I addressed the user requirements

As well as meeting the system requirements, I believe that I have met the user requirements for this program.

Firstly, I made it easy to use by continuing the existing menu system. The menus use coding so that the user only has to enter a single number to choose the option that they wish to see. The menus also provide helpful messages if the user enters an incorrect option.

Secondly, the code displays information in a meaningful way, I have used a mix of terminal output and the plotting functionality in matplotlib to provide graphs of the data. Providing data in both formats allows users to explore the data in a format that best suits their preferred style.

How the solution could be further developed

As I mentioned earlier in my review, there are some parts of the code that are a little inefficient. For example, the `currency_performance()` function has hard coded options for the three different currencies. If I was to develop this, I would add functionality that extracts the header rows from the csv file and uses this to both provide a list of options and retrieve the strings for passing to other functions. By doing this it would allow the code to be more scalable as it would be able to use a CSV file that contains a range of different currencies, providing a more useful tool for the company.

The second improvement I would make is to provide greater flexibility for the user when selecting the time frame they wish to analyse. I would do this by allowing the user to select the specific number of days that they wish to see. I think this would also allow me to reduce the amount of code in each of the main data processing functions as I could use a variable to store the number of days provided by the user and pass it to a single piece of code, rather than having 3 blocks of code that contain a lot of repetition.

Another option could be to allow the user to select a start and end date for the analysis, rather than by a number of days.

Task 4b - Commentary

The student has provided a number of value judgments as to the quality of their work and attempted, mostly successfully, to relate these to the requirements of the brief. The student provides suitable reasoning and supports their judgment with specific examples from their work.

The student provides an honest and balanced evaluation considering both what they have done well and areas that they have not done as well.

The student also provides a range of suggested improvements which are appropriate and meaningful, taking into consideration the requirements of the brief and what they have and haven't achieved.

At times, however, the evaluation becomes a little bit descriptive and misses the opportunity to evaluate some of the more technical aspects of their solution. For example, they could explore:

- why they have chosen a specific graph format and how that helps the user
- how they would implement the option to search by date and how that would impact on the performance of the code.

Overall, the student's work demonstrates a significant number of excellent aspects which would contribute to meeting the Grade A descriptor such as:

- The student was able to comprehensively review how well digital solutions meet a brief using reflective evaluation.
- They have supported their evaluation within depth understanding of the context and provided convincing rationales for improvement.
- Effective use of English skills to communicate to a technical audience.



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